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CSE-4E

DEVOPS

MSE2

Test Case 1 — Linux System Audit & User Management

Scenario: You are a Linux system administrator onboarding new employees at a tech company. Complete the following tasks on your system.

Tasks

1. Check the identity of the currently logged-in user and display the username of the current session.
2. Check which users are currently logged into the system and what they are doing.
3. View the login history of the system.
4. Display complete system information, the hostname, and how long the system has been running.
5. Create two groups: developers and qa. Verify both groups were successfully created and display the complete list of groups on the system.
6. Create the following users and assign each to their respective group:

Employee	Department
aman	developers
ritu	developers
karan	qa

7. Employee ritu has resigned. Ensure she has no active sessions, delete her account along with her home directory, and verify she no longer exists on the system.

Expected Outcome

- System audit commands display current user, active sessions, login history, hostname, and uptime without errors.
- Groups developers and qa appear in the system group list.
- Users aman and karan are created and assigned to the correct groups (verified via id command).
- After deletion, any attempt to look up ritu returns: **"no such user"**.

1. Check current logged-in user

whoami

2. Check currently logged-in users and their activity

who

Detailed info

w

3. View login history

last

4. Display system information, hostname, and uptime

System information

uname -a

Hostname

hostname

Uptime

uptime

5. Create groups: developers and qa

sudo groupadd developers

sudo groupadd qa

Verify groups created

getent group developers

getent group qa

Display all groups

cat /etc/group

6. Create users and assign to groups

Create aman in developers

```
sudo useradd -m -g developers aman

# Create ritu in developers
sudo useradd -m -g developers ritu

# Create karan in qa
sudo useradd -m -g qa karan
```

Verify user group assignment

```
id aman
id ritu
id karan
```

7. Remove user ritu after resignation

```
# Delete user along with home directory
sudo userdel -r ritu

# Verify deletion
id ritu
```

```
 ^Hgfgvanshika@LAPTOP-00FVIKGS:~$ whoami
vanshika
vanshika@LAPTOP-00FVIKGS:~$ id
uid=1000(vanshika) gid=1000(vanshika) groups=1000(vanshika),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plug
dev),100(users),107(netdev),1011(docker)
vanshika@LAPTOP-00FVIKGS:~$ who
vanshika pts/1        2026-04-24 04:43
vanshika@LAPTOP-00FVIKGS:~$ w
 04:44:05 up 1 min,  1 user,  load average: 0.13, 0.05, 0.02
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU  WHAT
vanshika pts/1    -                04:43   56.00s  0.07s  0.05s -bash
vanshika@LAPTOP-00FVIKGS:~$ last
reboot   system boot  6.6.87.2-microso Fri Apr 24 04:43   still running
reboot   system boot  6.6.87.2-microso Fri Mar 20 08:20 - 10:19  (01:58)
reboot   system boot  6.6.87.2-microso Thu Mar 19 10:20 - 10:20  (00:00)
reboot   system boot  6.6.87.2-microso Thu Mar 19 10:19 - 10:19  (00:00)
reboot   system boot  6.6.87.2-microso Thu Mar 19 10:19 - 10:19  (00:00)
reboot   system boot  6.6.87.2-microso Mon Feb 23 08:12 - 11:10  (02:57)
reboot   system boot  6.6.87.2-microso Fri Feb 20 08:48 - 09:35  (00:47)
reboot   system boot  6.6.87.2-microso Fri Feb 20 08:47 - 08:48  (00:00)
reboot   system boot  6.6.87.2-microso Fri Feb 20 08:46 - 08:47  (00:00)
reboot   system boot  6.6.87.2-microso Fri Feb 20 08:46 - 08:46  (00:00)
reboot   system boot  6.6.87.2-microso Fri Feb 20 08:45 - 08:46  (00:00)

wtmp begins Fri Feb 20 08:45:30 2026
vanshika@LAPTOP-00FVIKGS:~$ hostnamectl
 Static hostname: LAPTOP-00FVIKGS
       Icon name: computer-container
         Chassis: container □
      Machine ID: 018457a80ad34294ad4cc9d6efcc30f3
         Boot ID: 1bcd05006db64003a1011abf9be4d7ea
  Virtualization: wsl
Operating System: Ubuntu 24.04.1 LTS
          Kernel: Linux 6.6.87.2-microsoft-standard-WSL2
    Architecture: x86-64
vanshika@LAPTOP-00FVIKGS:~$ sudo groupadd developers
[sudo] password for vanshika:
vanshika@LAPTOP-00FVIKGS:~$ sudo groupadd qa
```

```
vanshika@LAPTOP-00FVIKGS:~$ sudo groupadd qa
vanshika@LAPTOP-00FVIKGS:~$ getent group developers
developers:x:1012:
vanshika@LAPTOP-00FVIKGS:~$ getent getent group qa
Unknown database: getent
Try `getent --help' or `getent --usage' for more information.
vanshika@LAPTOP-00FVIKGS:~$ getent group qa
qa:x:1013:
vanshika@LAPTOP-00FVIKGS:~$ sudo adduser aman
info: Adding user `aman' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `aman' (1014) ...
info: Adding new user `aman' (1014) with group `aman (1014)' ...
info: Creating home directory `/home/aman' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for aman
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `aman' to supplemental / extra groups `users' ...
info: Adding user `aman' to group `users' ...
vanshika@LAPTOP-00FVIKGS:~$ sudo adduser ritu
info: Adding user `ritu' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `ritu' (1015) ...
info: Adding new user `ritu' (1015) with group `ritu (1015)' ...
info: Creating home directory `/home/ritu' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
```

```
Changing the user information for ritu
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `ritu' to supplemental / extra groups `users' ...
info: Adding user `ritu' to group `users' ...
vanshika@LAPTOP-00FVIKGS:~$ sudo adduser karan
info: Adding user `karan' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `karan' (1016) ...
info: Adding new user `karan' (1016) with group `karan (1016)' ...
info: Creating home directory `/home/karan' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for karan
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `karan' to supplemental / extra groups `users' ...
info: Adding user `karan' to group `users' ...
vanshika@LAPTOP-00FVIKGS:~$ sudo usermod -aG developers aman
vanshika@LAPTOP-00FVIKGS:~$ sudo usermod -aG developers ritu
vanshika@LAPTOP-00FVIKGS:~$ sudo usermod -aG qa karan
vanshika@LAPTOP-00FVIKGS:~$ id aman
uid=1014(aman) gid=1014(aman) groups=1014(aman),100(users),1012(developers)
vanshika@LAPTOP-00FVIKGS:~$ id ritu
uid=1015(ritu) gid=1015(ritu) groups=1015(ritu),100(users),1012(developers)
vanshika@LAPTOP-00FVIKGS:~$ id karan
```

```
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `karan' to supplemental / extra groups `users' ...
info: Adding user `karan' to group `users' ...
vanshika@LAPTOP-00FVIKGS:~$ sudo usermod -aG developers aman
vanshika@LAPTOP-00FVIKGS:~$ sudo usermod -aG developers ritu
vanshika@LAPTOP-00FVIKGS:~$ sudo usermod -aG qa karan
vanshika@LAPTOP-00FVIKGS:~$ id aman
uid=1014(aman) gid=1014(aman) groups=1014(aman),100(users),1012(developers)
vanshika@LAPTOP-00FVIKGS:~$ id ritu
uid=1015(ritu) gid=1015(ritu) groups=1015(ritu),100(users),1012(developers)
vanshika@LAPTOP-00FVIKGS:~$ id karan
uid=1016(karan) gid=1016(karan) groups=1016(karan),100(users),1013(qa)
vanshika@LAPTOP-00FVIKGS:~$ who | grep ritu
vanshika@LAPTOP-00FVIKGS:~$ sudo deluser --remove-home ritu
info: Looking for files to backup/remove ...
info: Removing files ...
info: Removing crontab ...
info: Removing user `ritu' ...
vanshika@LAPTOP-00FVIKGS:~$ id ritu
id: 'ritu': no such user
vanshika@LAPTOP-00FVIKGS:~$
```

Test Case 2 — File Permission Configuration

Scenario: A web server requires a properly configured index.html file with specific ownership and access permissions.

Tasks

1. Navigate to your home directory and create a file named index.html.
2. Add any HTML text content to the file.
3. Check the file's current default permissions.
4. Grant full permissions (read, write, execute) to all users on the file.
5. Change the file owner and group to www-data.
6. Change the file permissions to 755.

7. Display the final permissions and ownership to verify the configuration.

Expected Outcome

- File index.html exists in the home directory with content.
- After step 4, permissions show rwxrwxrwx.
- After step 5, owner and group both show www-data.
- Final ls -l output shows:

-rwxr-xr-x 1 www-data www-data ... index.html

```
# 1. Navigate to home directory
cd ~

# 2. Create index.html file
touch index.html

# 3. Add HTML content to the file
echo "<html><body><h1>Hello World</h1></body></html>" > index.html

# 4. Check current default permissions
ls -l index.html

# 5. Grant full permissions to all users
chmod 777 index.html

# Verify permissions
ls -l index.html
```

6. Change owner and group to www-data

sudo chown www-data:www-data index.html

Verify ownership

ls -l index.html

7. Change permissions to 755

chmod 755 index.html

8. Display final permissions and ownership

ls -l index.html

```
vanshika@LAPTOP-00FVIKGS:~$ cd ~
vanshika@LAPTOP-00FVIKGS:~$ touch index.html
vanshika@LAPTOP-00FVIKGS:~$ nano index.html
vanshika@LAPTOP-00FVIKGS:~$ ls -l index.html
-rw-r--r-- 1 vanshika vanshika 60 Apr 24 05:14 index.html
vanshika@LAPTOP-00FVIKGS:~$ chmod 777 index.html
vanshika@LAPTOP-00FVIKGS:~$ ls -l index.html
-rwxrwxrwx 1 vanshika vanshika 60 Apr 24 05:14 index.html
vanshika@LAPTOP-00FVIKGS:~$ sudo chown www-data:www-data index.html
[sudo] password for vanshika:
vanshika@LAPTOP-00FVIKGS:~$ chmod 755 index.html
chmod: changing permissions of 'index.html': Operation not permitted
vanshika@LAPTOP-00FVIKGS:~$ sudo chmod 755 index.html
vanshika@LAPTOP-00FVIKGS:~$ ls -l index.html
-rwxr-xr-x 1 www-data www-data 60 Apr 24 05:14 index.html
vanshika@LAPTOP-00FVIKGS:~$ sudo chmod 755 index.html
vanshika@LAPTOP-00FVIKGS:~$
```

Test Case 3 — Node.js Application Setup & Git Integration

Scenario: You are setting up a new Node.js REST API from scratch and publishing it to GitHub.

Tasks

1. Initialize a new Node.js project and install the required dependencies: express and @types/express (as a dev dependency).
2. Create an index.js file that starts an Express server on port 8080 with the following endpoint:
    - GET /health — returns HTTP status 200 with the JSON response { "status": "OK" }.
3. Initialize a Git repository in the project folder.
4. Create an initial commit with all project files.
5. Create a GitHub repository named nodejs-api and push your code to it.

Expected Outcome

- Project folder contains package.json, node_modules/, and index.js.
- Running the server locally and calling GET /health returns:

{ "status": "OK" }

- GitHub repository nodejs-api is publicly visible with at least one commit containing all project files.

```
# ========================================
# NODE.JS APPLICATION SETUP & GIT INTEGRATION
# ========================================

# 1. Create project folder and navigate into it
mkdir nodejs-api
cd nodejs-api

# 2. Initialize Node.js project
npm init -y
```

```
# 3. Install dependencies
npm install express
npm install --save-dev @types/express

# 4. Create index.js file
touch index.js

# 5. Add Express server code
# Open index.js and paste the following:

const express = require('express');
const app = express();
const PORT = 8080;

app.get('/health', (req, res) => {
   res.status(200).json({ status: "OK" });
});

app.listen(PORT, () => {
   console.log(`Server running on http://localhost:${PORT}`);
});


# 6. Run the server
node index.js
```

```
# Test endpoint:
# Open browser or use curl:
# curl http://localhost:8080/health

# Expected Output:
# { "status": "OK" }


# ================================
# GIT SETUP
# ================================

# 7. Initialize Git repository
git init

# 8. Add all files
git add .

# 9. Commit files
git commit -m "Initial commit - Node.js API setup"


# ================================
# GITHUB INTEGRATION
# ================================
```

```
# 10. Create a repository on GitHub named: nodejs-api

# 11. Add remote origin (replace with your repo URL)
git remote add origin https://github.com/your-username/nodejs-api.git

# 12. Push code to GitHub
git branch -M main
git push -u origin main


# ================================
# EXPECTED OUTPUT
# ================================

# Files in project:
# package.json
# node_modules/
# index.js

# API Response:
# { "status": "OK" }

# GitHub repo:
# nodejs-api (public) with at least one commit
```

==========

END OF TASK

==========

```
vanshika@LAPTOP-00FVIKGS:~$ mkdir nodejs-api
vanshika@LAPTOP-00FVIKGS:~$ cd nodejs-api
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ npm init -y
Wrote to \\wsl.localhost\Ubuntu\home\vanshika\nodejs-api\package.json:

{
  "name": "nodejs-api",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}


vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ npm install express
npm install --save-dev @types/express

added 65 packages, and audited 66 packages in 5s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

added 11 packages, and audited 77 packages in 5s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ nano index.js
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ node index.js
Command 'node' not found, but can be installed with:
sudo apt install nodejs
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ sudo apt update
[sudo] password for vanshika:
```

```
Setting up node-assert (2.0.0+~cs3.9.8-2) ...
Setting up node-css-loader (6.8.1+~cs14.0.17-1) ...
Setting up node-parse-json (5.2.0+~cs5.1.7-1) ...
Setting up npm (9.2.0~ds1-2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ node -v
v18.19.1
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ npm -v
11.6.2
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ node index.js
Server running on http://localhost:8080

^C
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/vanshika/nodejs-api/.git/
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ ^C
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ nano .gitignore
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ git add .
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ git commit -m "Initial commit - Node API"
[master (root-commit) 4422e0e] Initial commit - Node API
 4 files changed, 966 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 index.js
 create mode 100644 package-lock.json
 create mode 100644 package.json
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ git branch -M main
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ git remote add origin https://github.com/vanshikaMudgal/nodejs-api.git
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$ git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 8.95 KiB | 2.98 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/vanshikaMudgal/nodejs-api.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
vanshika@LAPTOP-00FVIKGS:~/nodejs-api$
```